Language of Division

Speech Analysis of Far-Right Rhetoric in Germany—The AfD Case Study

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Abstract

Far-right parties have increasingly overtime posed harm to marginalized communities and heightened tensions with other nations. Over the past decade, both Europe and North America have witnessed a significant rise in far-right political movements. In Germany, the 2025 federal elections marked an unprecedented surge in support for the far-right party, Alternative für Deutschland (AfD). Electoral maps revealed a geographical split that mirrored the pre-reunification divide of Germany before 1990. This thesis investigates how the rhetoric of the AfD has contributed to this societal division. Using Artificial Intelligence, 1,912 speeches and interviews by party leaders and members of parliament were transcribed. The study applies Natural Language Processing (NLP) techniques to categorize far-right political discourse. Additionally, a predictive model was developed to assess the political leaning of any given speech, identifying its similarity to far-right rhetoric. The model successfully identifies right-wing texts and distigushes it from left-wing texts. This model functions as an early warning system to detect political actors who may adopt far-right ideologies and narratives. Analysis of the speeches revealed a consistent rhetorical pattern that constructs a binary between "you" as the in-group and "them" as the out-group.

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01. Introduction

Language is human's primary tool for communication, allowing individuals to convey information, express emotions, and negotiate relationships. From casual conversations and storytelling to the execution of professional tasks, language enables cooperation and the sharing of knowledge across generations. Beyond its communicative function, language shapes thought and reality. As the linguist Edward Sapir observed, "Language is a guide to social reality", it powerfully conditions how we perceive the world around us and even "the world of social activity" we take part in.[1] Psychologists similarly note that thought itself is largely mediated by language; for instance, Lev Vygotsky argued that thinking "comes into existence" through words rather than merely being clothed by them.[2]

Given its importance, it is no surprise that language can be used as a tool of influence and manipulation. The words and phrases people encounter can subtly frame their understanding of events, steering opinions and emotions often without conscious realization. Political propagandists, advertisers, and other persuaders have long exploited this power of language to shape public opinion. Classic examples range from euphemistic wartime propaganda to the spin of modern media messaging. Language not only reflects reality but actively constructs it in the minds of audiences. Scholars emphasize that language can serve as "a method of manipulation and control over mass consciousness," not merely a neutral vehicle of information exchange[3]. In political or media discourse, carefully chosen terms may amplify fear or hope, incite anger or sympathy, all in service of a manipulator's goals. As George Lakoff notes, language activates unconscious associations, a well-chosen phrase can cue a conceptual frame and thereby guide listeners toward particular conclusions.[4] Politicians, especially those with large audiences, often use emotions like fear, pride, or outrage in their speeches to encourage collective action or blame specific groups and create a division. Research into extremist rhetoric shows that certain words can heighten social tensions. For example, language that dehumanizes others or emphasizes conflicts as a battle between "us" and "them" creates an environment of hatred and division.

Politics is fundamentally carried out through language in public debates, manifestos, laws, news coverage, and recently mainly over social media posts, making discourse a primary medium through which power operates. Political language is persuasive: it is crafted to inform, misinform and convince, manipulate, and sometimes deceive. The discourse analyst Ruth Wodak noted that, political communication is characterized by the use of rhetorical and sometimes "delusive" devices; politicians across different formats employ a repetitive pattern of discursive strategies to achieve their goals.[5]

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[1] Edward Sapir, quoted in Karen J. Foli, "The Power of the Language We Use: Stigmatization of Individuals and Fellow Nurses with Substance Use Issues," *Research in Nursing & Health* 45, no. 6 (2022): 571.
[2] Lev S. Vygotsky, Thought and Language, edited and translated by Alex Kozulin (Cambridge, MA: MIT Press, 1986), 94.
[3] Teun A. van Dijk, *Ideology: A Multidisciplinary Approach* (London: SAGE Publications, 1998), 71.
[4] George Lakoff, *The All New Don't Think of an Elephant! Know Your Values and Frame the Debate* (White River Junction, VT: Chelsea Green, 2014), 4–6.
[5] Ruth Wodak, "Political Discourse Analysis – Theoretical and Methodological Challenges," in *Qualitative Discourse Analysis in the Social Sciences*, ed. Ruth Wodak and Michał Krzyżanowski (Basingstoke: Palgrave Macmillan, 2008), 6–7.

Through language, political actors construct narratives about societal problems and solutions, articulate values, and define group identities. In particular, the communication of ideology occurs largely via linguistic cues. Ideologies give distinct color to political language: for example, a socialist politician's speeches might emphasize "justice," "equality," and collective terms like "workers" or "community," whereas a nationalist or far-right politician might invoke "tradition," "sovereignty," and out-group labels like "foreigners" or "elites" in pejorative ways.

Teun van Dijk, in his studies of political ideology and discourse, highlights the pervasive "Us vs. Them" dichotomy: texts often emphasize the positive attributes of "us" (the speaker's group or those they claim to represent) while portraying "them" (political opponents, immigrants, minorities, etc.) in a negative light. The Alternative for Germany (AfD), a far-right populist party, exemplifies this rhetorical strategy. Its discourse frequently contrasts "the German people" "us" "you" with the established political elite in Berlin or with non-German outsiders "they", "them", suggesting that "you", "we" have been betrayed or endangered by "them". Through such narratives, AfD leaders propagate an ideology rooted in nativism and anti-elitism. After wining around 20% of the votes in the 2025 elections, the AfD contributed to a socio-political divide between East and West Germany, eerily reflecting the geographic and ideological division that existed before reunification in 1990. This division raises concerns about the growing influence of far-right politics and the possibility of history repeating itself in Germany.

The central question driving this thesis is: How does the language used by the AfD leading parliamentary members contribute to societal division in Germany? In particular, the project examines the rhetorical mechanisms through which AfD politicians may be exploiting and reinforcing the division that appeared after the 2025 elections, in their speeches and communications. The research approach is focusing on offical speeches by party leaders. A key source of data is a corpus of 1200 speeches delivered by AfD figures in the German Parliament official meetings that have been recorded and shared via the party official YouTube channel and addtionaly 800 speeches by the "Die Linke" the left party in Germany for comparison and machine learning training. The research is interested in What linguistic patterns define far-right discourse in Germany? And how can we deploy a Machine learning model to identify political parties that adopt far-right rhetoric without explicitly declaring their ideological stance?

02. Methodology

02.01. Natural Language Processing and classification

In recent years, the study of political language has been revolutionized by technological advancements in computational linguistics. Whereas traditionally a single researcher could only manually analyze a limited number of speeches or documents, now Natural Language Processing (NLP) techniques and machine learning models enable the systematic analysis of large textual corpora.

This “text-as-data” movement treats words as quantifiable data points, allowing political scientists and linguists to detect patterns and test hypotheses across thousands of texts with speed and rigor. Automated content analysis greatly lowers the cost and time of examining political communication on a large scale, unlocking possibilities that were previously impractical for all but the most well-funded projects.[6]

A wide array of NLP methods has been brought to bear on political texts. Some approaches use dictionaries or lexicons, predefined lists of words associated with particular categories (say, positive vs. negative tone, or liberal vs. conservative ideology), to measure the frequency of language indicating those categories. Other approaches employ machine learning to classify or cluster texts based on patterns the algorithms learn from data. For instance, researchers have trained classifiers to predict a document’s political affiliation or the speaker’s party from the language alone, with considerable success.[7] Such models, including supervised machine learning and even deep learning techniques like neural networks, can discern subtle differences in word choice and style that correlate with ideology. One study demonstrated that a neural network could infer political ideology from text by picking up on latent patterns in phrasing and syntax that escape superficial reading.[8] Likewise, computational models have been used to detect media bias in news articles by identifying tells in vocabulary and framing.[9] Beyond classification tasks, topic modeling has emerged as a powerful unsupervised tool to uncover the latent themes in a collection of documents.

Beyond classification tasks, topic modeling has emerged as a powerful unsupervised tool to uncover the latent themes in a collection of documents. By analyzing word co-occurrence, topic models can reveal, for example, which issues (immigration, economy, climate, etc.) dominate a party’s manifestos or how discourse priorities evolve over time. Applying such methods to party programs or campaign speeches allows analysts to quantify shifts in agenda or the prominence of ideological content. Studies of party manifestos using NLP have shown how parties re-calibrate their language across election cycles, emphasizing certain topics or adopting rivals’ rhetoric when strategic.[10]

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[6] Justin Grimmer and Brandon M. Stewart, “Text as Data: The Promise and Pitfalls of Automatic Content Analysis Methods for Political Texts,” *Political Analysis* 21, no. 3 (2013): 267–297.
[7] Mohit Iyyer et al., “Political Ideology Detection Using Recursive Neural Networks,” in *Proceedings of the 52nd Annual Meeting of the ACL (Volume 1)* (Baltimore, MD: ACL, 2014), 1113–1122.
[8] Iyyer et al., “Political Ideology Detection,” 1115.
[9] Ramy Baly et al., “We Can Detect Your Bias: Predicting the Political Ideology of News Articles,” *arXiv* (2020), https://arxiv.org/abs/2010.05338.
[10] Grimmer and Stewart, “Text as Data,” 284.

Another important set of techniques involves sentiment analysis, which automatically gauges the emotional tone of text. Sentiment analysis can measure positivity/negativity or more fine-grained affect, such as fear, anger, or joy expressed in political messaging. For example, Soroka and colleagues (2015) employed sentiment scoring on news content to compare levels of negativity, fear, and anger in media coverage.[11]

The use of NLP and machine learning in political discourse analysis is not without challenges. Algorithms may inadvertently carry biases or miss context that a human reader would catch; therefore, researchers stress the importance of careful validation and interpretability. Nonetheless, these technologies have greatly extended our capacity to analyze political language.

02.02. Data Collection and Processing

A fundamental question in data-driven research is: Where is the data? Before beginning any analysis, it is essential to determine whether relevant data exists, how it can be accessed, and how it should be structured to effectively address the research question. In this study, which focuses on the rhetoric of the far-right party Alternative for Germany (AfD), it was clear from the outset that working with primary data would be crucial. Collecting original data would allow for greater control over its structure, categorization, and quality.

The main source of this primary data was speeches of members of parliament from the AfD fraction in the german parliament (Bundestag), collected from the party's offical channel on youtube, where they regualry post their speeches.

These videos provided a valuable lens through which to examine the language strategies employed by the AfD. The process of collecting and preparing this data involved multiple stages, from scraping and downloading to transcription, translation, and thematic categorization.

The process of collecting and preparing data for this study began with scraping content from YouTube. I used Apify, a cloud-based web scraping platform that enabled efficient and structured retrieval of content. With Apify, I was able to extract essential metadata such as video descriptions, title and link.

With the metadata secured, the next step involved downloading the actual video and audio files. I used yt-dlp is a feature-rich command-line audio/video downloader with support for thousands of sites. The project is a fork of youtube-dl based on the now inactive youtube-dlc.[12] The process was smooth and allowed for easy integration with the subsequent steps of my workflow (see Technical Appendix for implementation details). Yt-dlp helped downloading the audio seperatly which I was looking for to transcript and get the text later to start the analysis.

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[11] Stuart Soroka, Lauren G. Young, and Meital Balmas, "Bad News or Mad News? Sentiment Scoring of Negativity, Fear, and Anger in News Content," Annals of the American Academy of Political and Social Science 659, no. 1 (2015): 108–121.
[12] yt-dlp contributors, *yt-dlp*, GitHub repository, accessed May 10, 2025, https://github.com/yt-dlp/yt-dlp.

Once the audio files were ready, I proceeded to transcribe the content using OpenAI's Whisper model. Whisper is a powerful multilingual automatic speech recognition (ASR) system trained on diverse datasets, and it offered excellent accuracy for German-language transcription. All transcribed texts were compiled and stored in JSON format, ready for the next stage of processing.

Since all the speeches were delivered entirely in German, translation became a critical step in order to conduct in-depth analysis in English. I tested several translation APIs, including OpenAI's Whisper (which also supports basic translation) and DeepL, a highly regarded platform for context-sensitive language translation. Ultimately, DeepL was selected for its superior contextual accuracy and nuanced handling of political language, which Whisper lacked. I used DeepL's free API, which allowed for up to 500,000 translated words per month, though I noted that future scaling of this project would require a paid subscription due to API limitations.

However, the translation process introduced a new challenge. When handling large amounts of text, DeepL occasionally failed to translate entire rows. To address this issue, I implemented a troubleshooting workflow using Python's langdetect library. This tool helped identify which rows remained untranslated by detecting their language. I then created a secondary function to selectively re-translate only the untranslated texts. By isolating and processing these specific rows, I was able to successfully produce a fully translated dataset, ensuring the accuracy and completeness of the textual data used for further linguistic analysis.

02.03. Appyling NLP on Collected data

To analyze the political rhetoric of the Alternative für Deutschland (AfD), I systematically examined two hundred speeches obtained directly from the party's official communication channels. These speeches encompass a diverse range of media formats, including parliamentary broadcasts, official statements delivered during regular Bundestag sessions, and interviews featuring prominent parliamentary members. This comprehensive dataset provides insight into the linguistic patterns and rhetorical strategies employed by AfD representatives in both formal and informal contexts. The analysis was primarily conducted using Python's Natural Language Toolkit (NLTK), a powerful library specialized in textual processing, enabling detailed linguistic and semantic exploration of the collected speeches.

In order to systematically extract meaningful insights from the collected speeches, I conducted multiple exploratory analyses aimed at capturing key rhetorical patterns and build thematic categorizations. Among these experiments, three approaches proved most effective and were ultimately adopted for detailed analysis.

First, I categorized words based on their use in constructing an "us versus them" narrative, identifying terms that delineate clear in-groups ("you") and out-groups ("them").

Second, exploring main themes and main categories, like Enironment for example and the words associated with this topic.

Third, I implemented a general sentiment analysis to quantitatively assess the emotional tone (positive, neutral, or negative) within the speeches.

These combined methods provided a framework for understanding both the limitations and the approaches to analyzing natural language, enabling a more accurate and unbiased classification of political speeches.

Before diving into each analysis group, I want to highlight the initial general steps taken for text preprocessing. The first step involved converting all words to lowercase. This ensures consistency across the speeches by treating identical words the same, regardless of their original casing. Without this step, words like “Speech” and “speech” would be considered distinct.

The second step was the removal of punctuation, implemented using the (string.punctuation) function provided by Python's NLTK library.

The third step involved tokenizing each speech, converting them into lists of individual tokens (words), and subsequently removing basic stop words using NLTK's built-in stop word list.

Basic stop words:

[a', 'about', 'above', 'after', 'again', 'against', 'ain', 'all', 'am', 'an', 'and', 'any', 'are', 'aren', "aren't", 'as', 'at', 'be', 'because', 'been', 'before', 'being', 'below', 'between', 'both', 'but', 'by', 'can', 'couldn', "couldn't", 'd', 'did', 'didn', "didn't", 'do', 'does', 'doesn', "doesn't", 'doing', 'don', "don't", 'down', 'during', 'each', 'few', 'for', 'from', 'further', 'had', 'hadn', "hadn't", 'has', 'hasn', "hasn't", 'have', 'haven', "haven't", 'having', 'he', "he'd", "he'll", "he's", 'her', 'here', 'hers', 'herself', 'him', 'himself', 'his', 'how', 'i', "i'd", "i'll", "i'm", "i've", 'if', 'in', 'into', 'is', 'isn', "isn't", 'it', "it'd", "it'll", "it's", 'its', 'itself', 'just', 'll', 'm', 'ma', 'me', 'mightn', "mightn't", 'more', 'most', 'mustn', "mustn't", 'my', 'myself', 'needn', "needn't", 'no', 'nor', 'not', 'now', 'o', 'of', 'off', 'on', 'once', 'only', 'or', 'other', 'our', 'ours', 'ourselves', 'out', 'over', 'own', 're', 's', 'same', 'shan', "shan't", 'she', "she'd", "she'll", "she's", 'should', "should've", 'shouldn', "shouldn't", 'so', 'some', 'such', 't', 'than', 'that', "that'll", 'the', 'their', 'theirs', 'them', 'themselves', 'then', 'there', 'these', 'they', "they'd", "they'll", "they're", "they've", 'this', 'those', 'through', 'to', 'too', 'under', 'until', 'up', 've', 'very', 'was', 'wasn', "wasn't", 'we', "we'd", "we'll", "we're", "we've", 'were', 'weren', "weren't", 'what', 'when', 'where', 'which', 'while', 'who', 'whom', 'why', 'will', 'with', 'won', "won't", 'wouldn', "wouldn't", 'y', 'you', "you'd", "you'll", "you're", "you've", 'your', 'yours', 'yourself', 'yourselves']

After analyzing the speeches, I added additional words to the stop words list. These words did not contribute meaningful information to the analysis and instead introduced confusion. They can be thought of as category-related terms rather than informative keywords.

1. Numbers as words
   ['one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine', 'ten','first', 'second', 'third', 'fourth', 'fifth']

2. Formal addresses
   ['mr', 'mrs', 'ms', 'dr', 'professor', 'madam', 'sir', 'ladies', 'gentlemen', 'colleagues', 'honorable', 'excellency','chancellor', 'president', 'minister', 'chairman', 'chairwoman']

3. Political procedure words
   ['motion', 'amendment', 'vote', 'debate', 'session', 'agenda', 'parliament', 'congress', 'committee', 'subcommittee', 'hearing', 'speaker', 'chair', 'floor', 'order', 'point', 'yield']

4. Time references
   ['today', 'yesterday', 'tomorrow', 'week', 'month', 'year', 'decade', 'monday', 'tuesday', 'wednesday', 'thursday', 'friday', 'saturday', 'sunday', 'january', 'february', 'march', 'april', 'may', 'june', 'july', 'august', 'september', 'october', 'november', 'december']

5. Common political filler words
   ['like', 'well', 'just', 'going', 'come', 'say', 'said', 'saying', 'want', 'wanted', 'take', 'make', 'making', 'made', 'put', 'think', 'thought', 'believe', 'look', 'see', 'get', 'got', 'getting', 'thank', 'thanks', 'please', 'yes', 'no', 'okay', 'applause']

6. General words with high frequncies and less meaning
['let', 'us', 'must', 'need', 'needs', 'would', 'could', 'should', 'fellow', 'citizens', 'nation', 'country', 'government', 'people', 'public', 'policy', 'issue', 'issues', 'question', 'questions', 'statement', 'position', 'fact', 'facts', 'matter', 'matters', 'thing', 'things', 'way', 'ways', 'that's']

Before filtering out the stop words, the dataset contained 336,226 tokens (words). After preprocessing and removing stop words, the number of tokens was reduced to 124,470, which served as the basis for the final analysis.

02.03.01 "You" and or vs. "Them"

To systematically analyze the rhetoric employed in the AfD speeches, the first analysis conducted was the "You and Them" grouping, aimed at identifying linguistic patterns of group differentiation. Words were categorized into "You" and "They" groups, each subdivided into thematic categories including national, racial/ethnic, religious, cultural, and ideological/political terms. This categorization allowed a detailed exploration of how AfD members linguistically construct in-groups and out-groups.

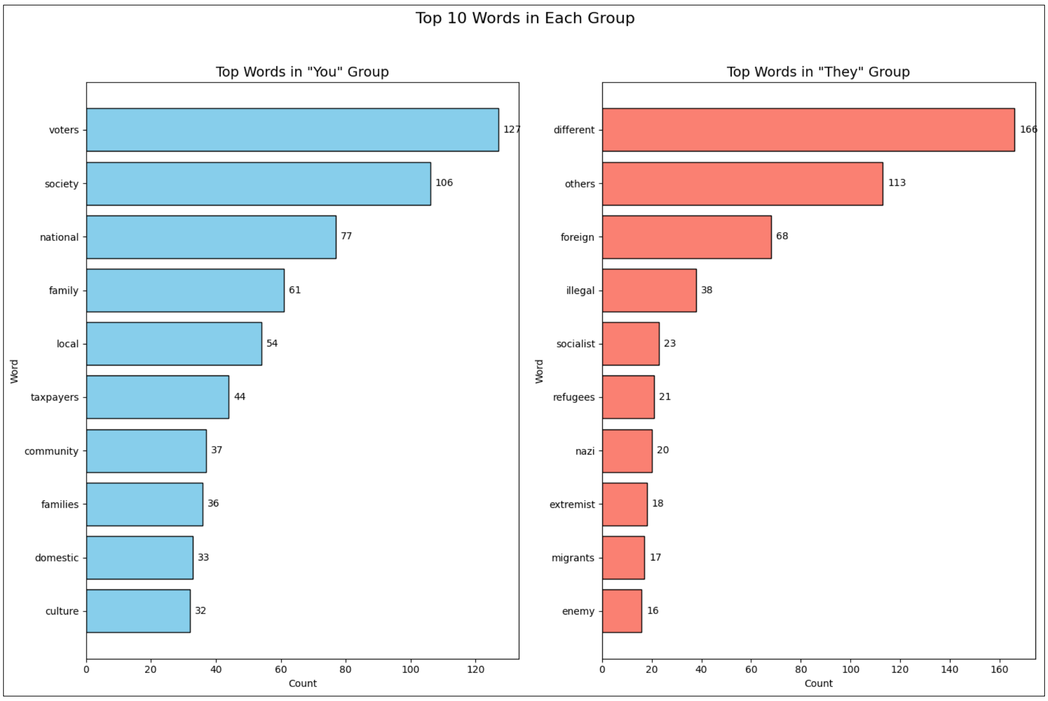

Figure 01. Top 10 words in both "You" and "They" groups

The categorization produced 92 distinct matching terms, with 53 words identified in the "You" group and 39 in the "They" group. The top 10 frequently occurring words from each group revealed significant insights into AfD's rhetorical strategies. The most prevalent words within the "You" group included terms like "party," "voters," "society," "national," and "family," highlighting a strong ideological and national identity framing. In contrast, the "They" group prominently featured words such as "different", "others," "foreign," and "illegal," indicating clear attempts to delineate outsiders or opponents, whether ethnically, culturally, or politically defined.

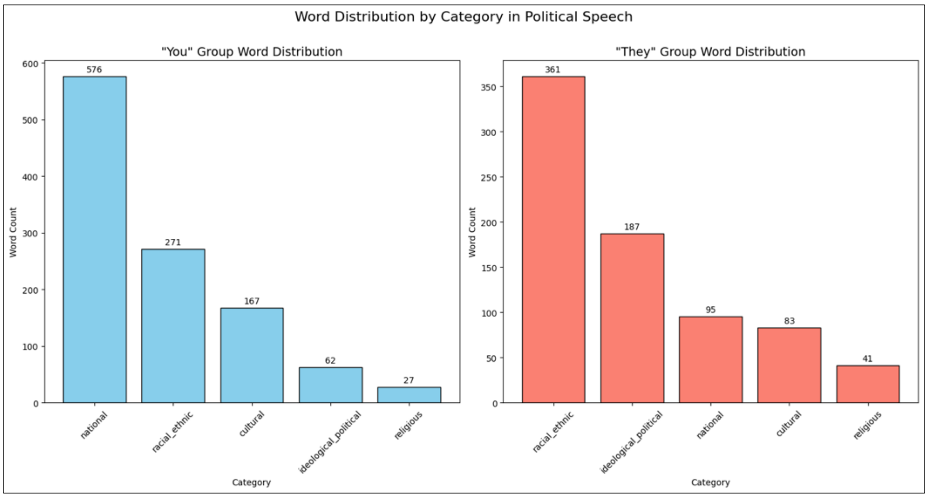

Figure 02. Word Counts across themes

02.03.02. Key Themes and Topics

In this section, the analysis focuses on identifying key themes and topics within the AfD speeches. To systematically capture the prominent subjects discussed by AfD members, I created thematic categories derived from typical political discourse areas. Each category includes words that reflect specific themes such as Identity, Economics, Politics, Security, Values, Environment, and Health & Social issues. By categorizing the speeches according to these thematic groups, it was possible to quantify and interpret the AfD's emphasis on particular policy areas and ideological positions. The categorization facilitated a structured examination of how frequently and intensively specific topics appeared across the speeches.

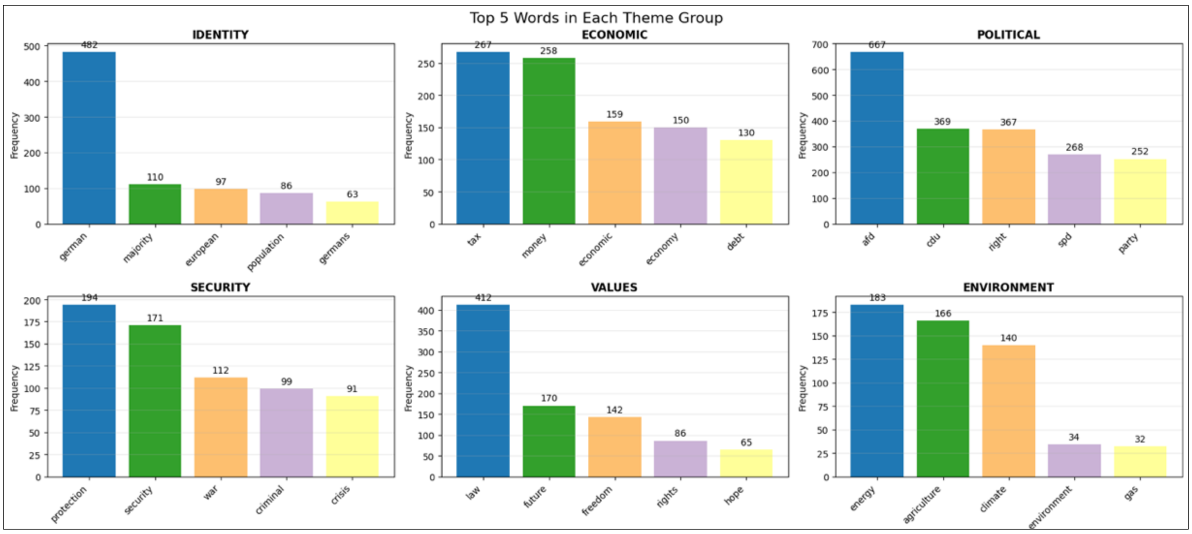

Figure 03. Words Count across all themes

02.03.03. Sentiment Analysis

In the next step of the analysis, I performed sentiment analysis within each thematic category using spaCy, a powerful open-source Python library designed for advanced Natural Language Processing (NLP). spaCy is particularly effective for sentiment analysis tasks because it efficiently processes large volumes of text, provides robust linguistic annotations, and supports advanced methods for identifying the emotional tone and sentiment polarity of texts. This capability enabled a nuanced assessment of how AfD's discourse emotionally frames different political themes.[13]

Overall, the AfD exhibits a predominantly negative tone in their speeches. The sentiment analysis was conducted across all thematic categories: Identity, Economic, Political, Values, and Environment. Although the results within each category vary, the aggregate sentiment remains negative. It is important to note that while sentiment analysis provides useful initial insights, it may not always accurately represent the underlying tone. For instance, a speech classified as hate speech might contain more positive words than negative ones, yet still convey an overwhelmingly negative message due to the intensity of certain negative terms. Therefore, employing machine learning methods for speech classification can offer a more accurate understanding of the overall patterns in political rhetoric for a specific party or individual. However, such an approach requires a substantial amount of data to build a robust classification model.

The sentiment within the Political category was strongly negative overall. It is important to note here that the word "party" may sometimes be classified incorrectly as positive, as it commonly refers to a celebration or festive event rather than a political party. The interpretation of "party" as referring to a political organization does not inherently carry positive or negative sentiment; it is inherently neutral. Therefore, analyzing the words immediately surrounding "party" can offer better contextual clues for sentiment classification. A bar chart illustrating the sentiments within this category is presented below.

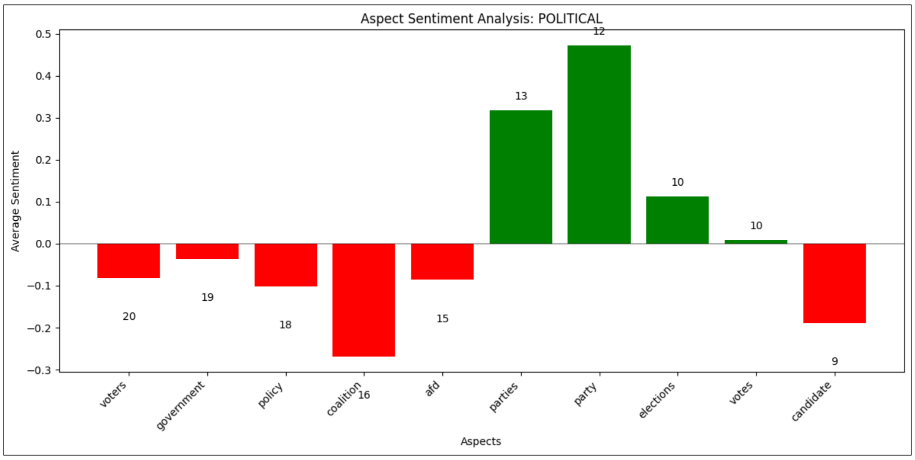

Figure 04. Word Counts and Their Sentiment in the Political Category

On the other hand, the Values category had a significantly higher count of positive words compared to negative ones. However, it's important to note that such analysis may not fully capture the true sentiment of the speech itself, but rather reflects the sentiment associated with individual words. For instance, words like “law” are generally classified as positive, yet they can appear in highly negative contexts, such as when a party discusses implementing a new law that targets a minority group.

[13] Explosion AI, *spaCy: Industrial-strength Natural Language Processing in Python* (2024), https://spacy.io/.

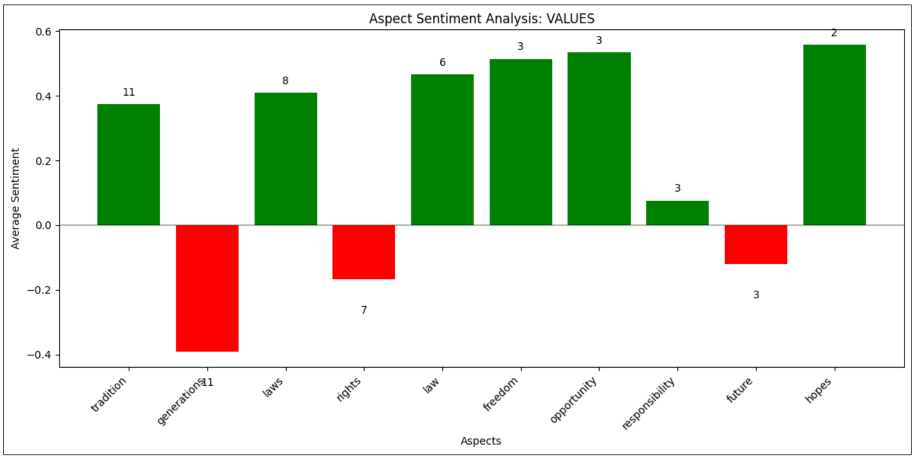

Figure 05. Word Counts and Their Sentiment in the Political Category

02.04. Visualization

After testing different chart types I decided on using bubbel charts to visualize the word frequencies across all speechs. The bubbel chart was effective because it's visualy powerfull for showing clusters and groups by just using distance. Size is very useful for represeting count and the number of circles was helpful to represent unique words. As well as the hirarchial stacked bubbel chart was particulary good for visualizing themes and the words within each theme.

To show the results of the sentiment analysis I used bidirectional barchart. Bidirectional bar chart is used for comparing two sets of data side by side along a vertical axis, which was exactly good to compare the sentiment analysis chart.

02.05. Machine Learning Model

A crucial part of this project involved training a machine learning model to assign a far-rightness score to any political speech. To frame this as a binary classification task, determining whether a speech was delivered by the far-right AfD, I needed a contrasting corpus. Therefore, I transcribed 800 speeches by Die Linke, Germany's left-wing party, so that the model could learn both the similarities and differences between left- and right-wing rhetoric and distinguish right-wing language from other forms. It is worth noting that while using only two parties is a reasonable starting point, a more generalized and robust model would require data from additional parties and a larger number of speeches. Another step toward improving the model would be developing a multilingual version that includes political speeches from parties across different countries. Due to time constraints and resource limitations, this project focused exclusively on German-language speeches from just one left-wing and one right-wing party.

I built the model with Gensim, an open-source Python library for scalable topic modelling and vector-space representations.[14] Specifically, I used Gensim's implementation of Doc2Vec, first introduced by Le and Mikolov in 2014.[15] Doc2Vec turns each document, in

[14] Radim Řehůřek and Petr Sojka, "Software Framework for Topic Modelling with Large Corpora," *Proceedings of the LREC 2010 Workshop on New Challenges for NLP Frameworks*, 2010
[15] Quoc V. Le and Tomas Mikolov, "Distributed Representations of Sentences and Documents," arXiv preprint arXiv:1405.4053, 2014

this case, an entire speech, into a dense vector. In Gensim terminology, the collection of these vectors is called a *corpus*.

A corpus in Gensim serves two main purposes: (1) as training input, allowing the algorithm to learn latent structures, and (2) as a basis for inference, enabling the model to generate vector representations, and thus topical information, for new, unseen documents. The "infer_vector" function is used to generate a vector for a new document based on a pre-trained model. This allows us to evaluate the new document and assess its similarity to the documents in the original training corpus.

Previous research shows that Doc2Vec can capture ideological signals in text and has been used to measure political bias in newspapers[16] and to classify the ideological direction of U.S. judicial opinions.[17] More recent work applies Doc2Vec-style embeddings to detect multiple forms of online extremism, demonstrating the method's suitability for far-right speech detection.[18] My corpus combines left- and right-wing speeches in a single JSON file. Each element is an object with two keys:

party (the label "Die Linke" or "AfD") and speech (the full transcript). The labelled corpus contains 1,912 transcripts.

At a high level, the workflow in a Machine Learning project is:

1. Problem definition (Classification, Regression, Clustering).
2. Data loading.
3. Data exploration.
4. Pre-processing.
5. Data split. Partition into training and test sets.
6. Model fitting. The model depends on the problem.
7. Evaluation. Assess accuracy.

In my project, I conducted several rounds of training as follows:

02.05.01 First round of training

The first round included all 1,912 speeches without any trimming—each speech was used as-is. To gain a detailed overview of the dataset, I ran descriptive statistics on the speech lengths to understand the characteristics of the training corpus.

[16] Hyungsuc Kang and Janghoon Yang, "Quantifying Perceived Political Bias of Newspapers through a Document Classification Technique," *Journal of Quantitative Linguistics* 29, no. 2 (2022): 127–50

[17] Carina I. Hausladen, Marcel H. Schubert, and Elliott Ash, "Text Classification of Ideological Direction in Judicial Opinions," *International Review of Law and Economics* 62 (2020): 105903.

[18] Mayur Gaikwad et al., "Multi-Ideology, Multiclass Online Extremism Dataset, and Its Evaluation Using Machine Learning," *Computational Intelligence and Neuroscience* (2023).

Charatcters descriptive statistics:

Metric	Value
Count (docuemnts)	1912
Mean (characters)	9705.84
Std Dev (characters)	17766.66
Min (characters)	76.00
25% (characters)	2972.00
50% (characters)	4397.00
75% (characters)	8409.50
Max (characters)	263360.00

Table 01. Characters descriptive statistics

Words descriptive statistics:

Metric	Value
Count (docuemnts)	1912
Mean (Words)	1498.29
Std Dev (Words)	2811.99
Min (Words)	11.00
25% (Words)	432.75
50% (Words)	640.50
75% (Words)	1275.00
Max (Words)	41797.00

Table 02. Words descriptive statistics

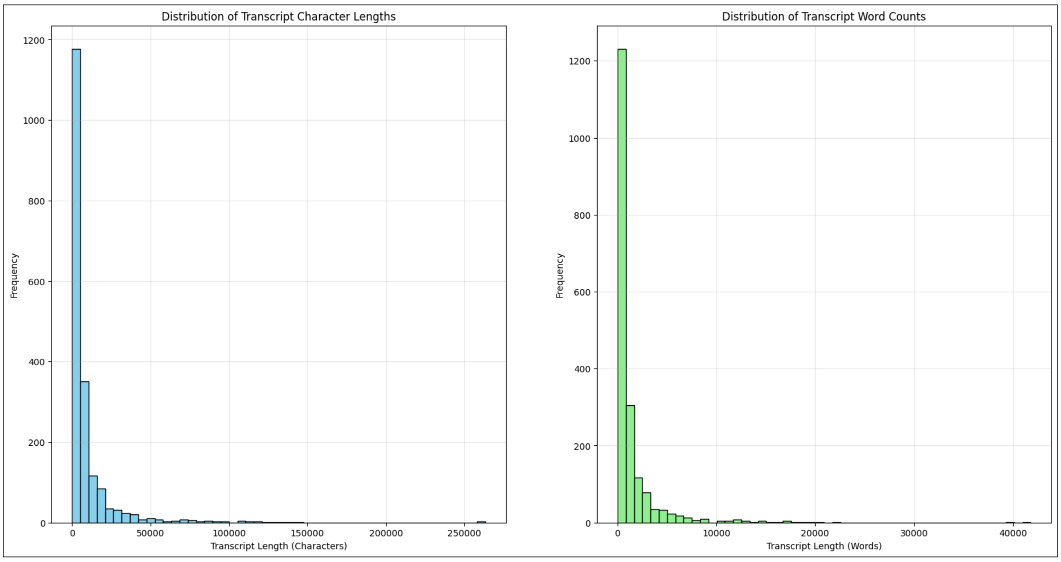

Figure 06. Characters and words descriptive statistics plots

To train a Doc2Vec model, we first need to tokenize the documents and tag them using the tagging system provided by the Gensim library. For the initial training round, I used a simple tokenization approach that retained the full content of each document without any preprocessing, using the NLTK library. Each document was then tagged with its corresponding party label (left- or right-wing) using the TaggedDocument function from Gensim. Next, I split the data into training and test sets, allocating 80% for training and 20% for testing. The model was then trained using basic default parameters.

```
vector_size=256,
window=8,
min_count=5,
workers=4,
sample=1e-4,
negative=5,
alpha=0.05,
min_alpha=0.001
```

These are basic parameters, similar to the default settings used to train a Doc2Vec model. After training the model for 20 epochs, the alpha value gradually decreased from 0.05 to 0.0035.

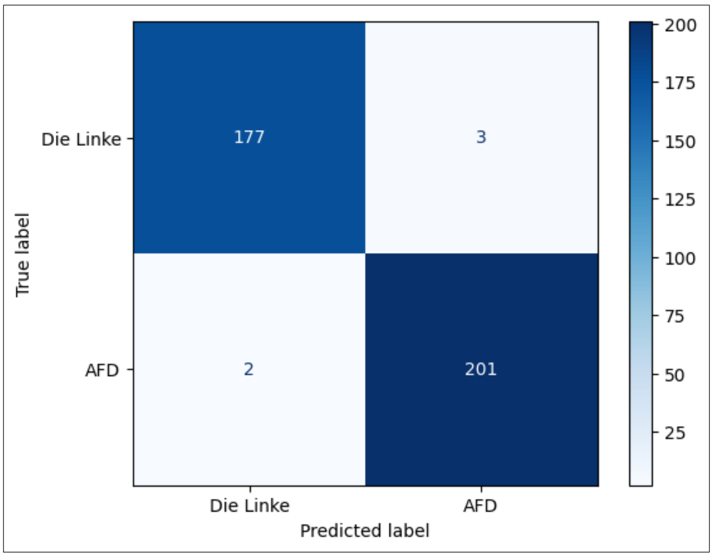

Figure 07. First model Confusion Matrix – Prediction resutls on test data

The Doc2Vec similarity classifier is a custom k-nearest neighbors approach I developed by identifying the top 10 most similar vector neighbors—vector representations of documents—and assigning a label based on the highest aggregated similarity score for each party. For example, if we input a single document and measure its similarity to the training dataset using Doc2Vec's .most_similar function, we might get a result like:

most_similar = ('train_1214', 0.5081)

We then extract the ID—1214 in this case—and reference the training dataset to determine the true label, which might be "Die Linke," the left-wing party. To avoid classification based on a single potentially noisy match, I applied the parameter topn=10 to retrieve the top 10 most similar documents. I then summed the similarity scores for each party label, producing results such as:

{ 'Die Linke': 3.76, 'AfD': 0.90 }

Based on this aggregate score, the speech can be confidently classified as a left-wing speech. This method was applied to the entire test dataset, and the resulting classifications were used to generate the confusion matrix shown above.

To evaluate how well the model performed, I used the classification_report function from the scikit-learn library. The results show that the Doc2Vec-based similarity approach works very well for classifying political speeches. It clearly picks up on the different ways each party uses language and is able to apply that knowledge to new, unseen speeches. This suggests that the model could be useful as an early-warning tool, helping us spot when a political speech starts to lean toward far-right or left-wing rhetoric, even if it's not obvious at first glance.

Classification Report for Doc2Vec Similarity Classifier

Class	Precision	Recall	F1-Score	Support
Die Linke (left-wing)	0.99	0.99	0.99	203
AfD (right-wing)	0.99	0.98	0.99	180
Accuracy			0.99	383
Macro Avg	0.99	0.99	0.99	383
Weighted Avg	0.99	0.99	0.99	383

Table 03. First model Accuracy scores

The performance of the Doc2Vec similarity classifier was evaluated using standard classification metrics, precision, recall, F1-score, and accuracy, on a test set of 383 political speeches. The results are summarized in Table above.

The classifier achieved 99% precision and recall for Die Linke (the left-wing party) and 99% precision and 98% recall for AfD (the far-right party). This indicates that the model is highly effective in both correctly identifying true positives and minimizing false positives across both categories.

The F1-score, which balances precision and recall, was 0.99 for both parties, suggesting a strong overall performance and a balanced ability to detect both left- and right-leaning speeches with high reliability.

The classifier correctly labeled 99% of the test samples, indicating a very high agreement between predicted and actual labels.

02.05.02 Second round of training

In the second round I have changed the legnths of all speechs to have each of a maximum number of 600 words. The porpus of this step is mainly for two reasons:

1. To check if the model would perform better when the speeches are less dense.
2. To have more documents to train the model on. Before reducing the length of each document I had 1912 documents in total after changing each document legnth I had 5709 docuemnts.

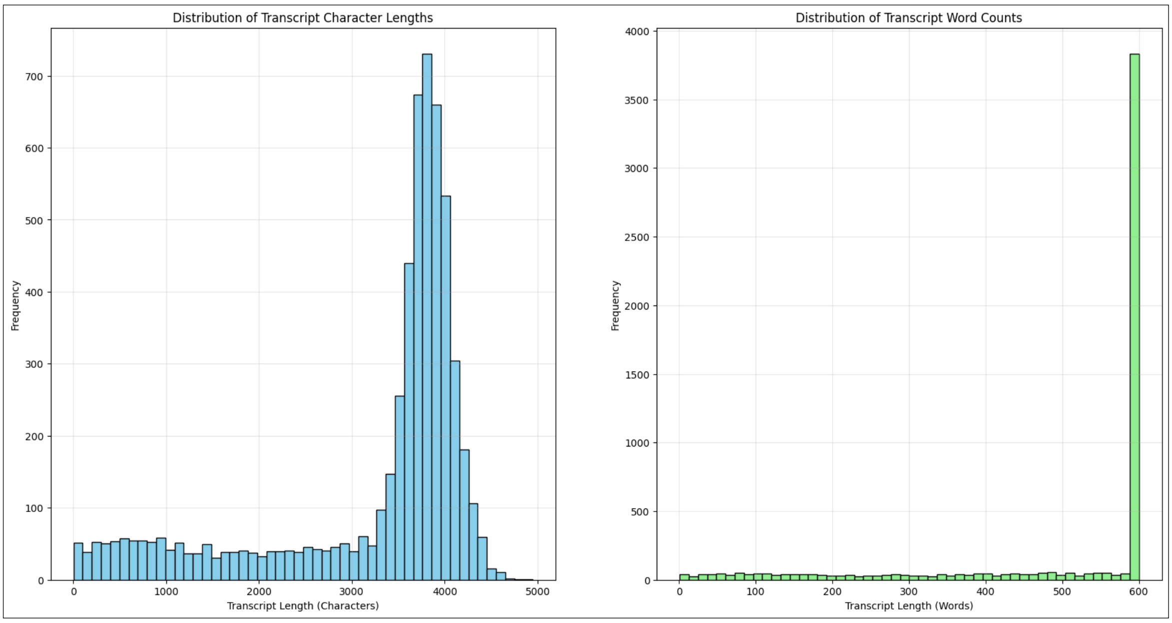

Figure 08. Plots showing number of characters across documents and number of words.

For a better overview of the data I ran a descriptive statistics function for a closer look after transforming the data:

Statistic	Character Length	Word Count
Count (new all documents)	5709	5709
Mean	3249.88	501.79
Std	1121.61	172.74
Min	5	1
25%	3136	468
50%	3728	600
75%	3921	600
Max	4947	600

Table 04. Character Length and Word Counts Descriptive Statistics

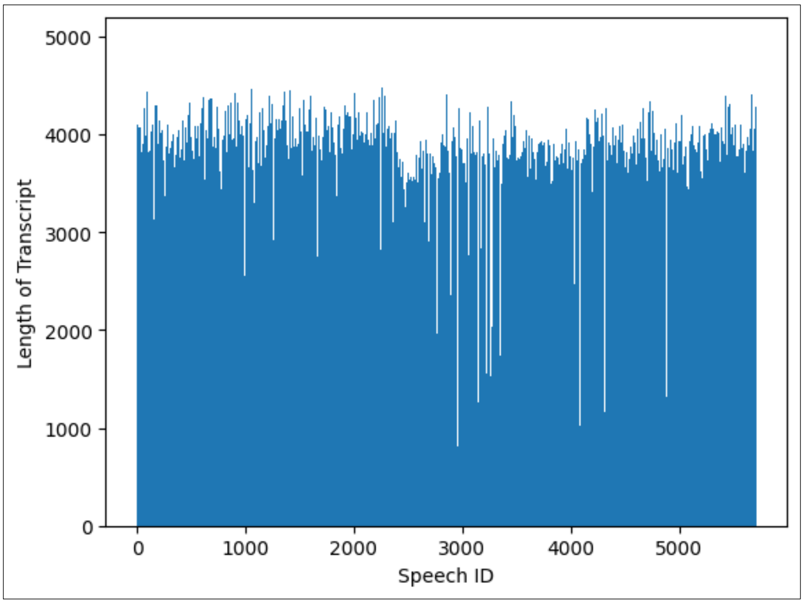

Figure 09. Number of Words Across all and Each Documents.

After reorganizing the documnts with new lengths, the data was splited into training and test datasets again using 80% (4564 documents) for training and 20% (1141 documents) for testing. The next step was to remove standard German stop words provided by NTLK the total number of removed stop words as unique words is 232 words. The total number of words in the dataset is 2,290,247 tokens in the training dataset before removing the stop words and 1,229,831 tokens after removing the stop words. The model was traind with the following specifications that were changed to perform the best training results givien the size of the dataset.

1. Sets `dm=0`, which sets the training algorithm to PV-DBOW and set `dbow_words=1` to update Word vectors. [More here](https://radimrehurek.com/gensim/models/doc2vec.html)

2. `vecor_size=100` is reduced to 100 because we have a small dataset. (The bigger the dataset the higher the number) -> the smallest number for the best result.

3. Increased from `window=8` to `window=10` to get broader topics within each speech. This parameter work together with the Document Tag

4. Setting `Sample=1e-4` as I have 1+ million tokens

5. Added `hs=0`Negative Sampling (NS) which works better, when `negative > 0`, better on low-medium corpa.

After finihsing the training the model was tested on the test dataset with the same costume nearest neighbor function developed in the first training phase as the data structure is the same.

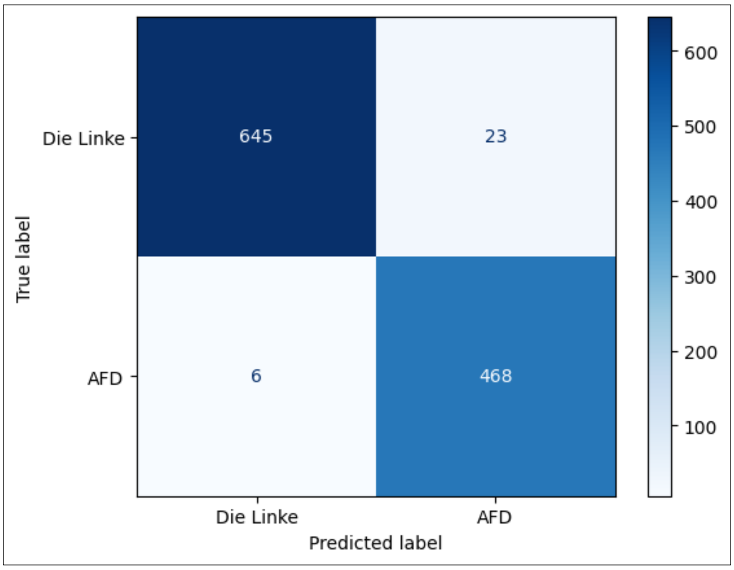

Figure 10. Model Results in Test data Confusion matrix (Die Linke: left wing party, AFD: the right-wing party)

Classification Report for Doc2Vec Similarity Classifier

Class	Precision	Recall	F1-Score	Support
Die Linke	0.95	0.99	0.97	474
AfD	0.99	0.97	0.98	668
Accuracy			**0.97**	1142
Macro Avg	0.97	0.98	0.97	1142
Weighted Avg	0.98	0.97	0.97	1142

Table 05. Model Evaluation Table

The performance of the Doc2Vec similarity classifier was evaluated using standard classification metrics: precision, recall, F1-score, and accuracy, based on a test set of 1,142 political speeches. The results are summarized in the table above. The classifier achieved a precision of 95% and recall of 99% for Die Linke (the left-wing party), and a precision of 99% and recall of 97% for AfD (the far-right party). This demonstrates the model's ability to distinguish between the two political orientations with high accuracy and minimal misclassification. The F1-scores, 0.97 for Die Linke and 0.98 for AfD, showes a strong and balanced performance across both classes. The overall accuracy of 97% further reflects the model's reliability in correctly identifying the political leaning of unseen speeches. However this is 2 percentage points less than the privouse model.

According to the results, I have chosen to integrate the first model at this point and continue training it to check for improved performance.

02.05.03 Model VueJs Integration

Once the Doc2Vec model had reached an acceptable level of performance, the next step was to make it accessible to end-users. Gensim's save() and load() methods allowed me to serialize the trained model to disk and reload it whenever needed. I then wrapped the model in a lightweight Gradio interface written in Python. Gradio is an open-source toolkit that can spin up a full REST-style endpoint and browser UI with only a few lines of code, making rapid prototyping straightforward.

The interface, the model weights, and a small test dataset were deployed to Hugging Face Spaces, which provides free hosting for machine-learning demos. Gradio also exposes a Python Client API that lets other Python scripts call the hosted model programmatically. My front-end, however, is built with Vue.js—a JavaScript framework that cannot import the Python client directly. Instead, Vue communicates with the Gradio back-end via standard HTTP requests (fetch/axios) to the auto-generated REST endpoints. While this approach keeps the tech stack clean and language-agnostic, it does require writing additional request-handling code on the Vue side, making the integration slightly slower to develop than the one-line Python client call.

03. Conclusion

Language is a powerful tool, it can build communities or foster division. It is the primary medium through which we connect, communicate, and shape our societies. Far-right parties have increasingly used language to polarize populations, creating internal divisions and international tensions. These parties often rely on similar rhetorical strategies across different national contexts.

In this project, I applied Natural Language Processing (NLP) techniques to analyze the speeches of Germany's far-right party, Alternative für Deutschland (AfD), to study how their rhetoric contributes to social division. The analysis revealed a recurring linguistic pattern that constructs a dichotomy between an in-group, addressed as "you" or "the people", and an out-group. often described with terms like "migrants," "criminal," or "dangerous." This rhetorical framing positions immigrants and outsiders as the source of societal problems, fostering exclusion and reinforcing nationalist narratives.

To deepen this analysis, I employed a machine learning approach using Document to Vector (Doc2Vec) to capture the structural and semantic similarities across over 1,900 speeches. The resulting classification model achieved 92% accuracy in distinguishing between left-wing and right-wing rhetoric, demonstrating the effectiveness of computational methods in identifying political language patterns.

This research not only sheds light on the divisive strategies of far-right discourse but also introduces a tool that could serve as an early-warning system for detecting the spread of extremist language in political communication.

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